

Topology-Controlled Reactive Mass Transfer in Biological Aggregates

Particle-Resolved Simulation, Graph-Based Model Reduction and Dynamic Closures for Reactor-Scale CFD–PBM Models

Revised Concept Summary

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Abstract

The project will determine how the internal topology of biological aggregates controls steady and transient oxygen supply and will translate this information into closures for reactor-scale simulations. Particle-resolved DNS–DEM simulations will resolve flow and oxygen transport around the individual cells of one aggregate at a time. A graph neural network (GNN) will be used *offline* as a structure-aware teacher model: it will test which topological information is required beyond size and porosity and will encode that information in a low-dimensional morphology representation. The GNN will not be evaluated for every aggregate in the final reactor simulation. Instead, its predictions will be compressed into a lookup table, symbolic correlation or small neural network for three quantities: equilibrium oxygen uptake, the dynamic response time and the fraction of poorly supplied cells. These deployable closures will be evaluated locally for the aggregate population in a CFD–population balance model (CFD–PBM) or for parcels in an Euler–Lagrange model. The project thereby connects cell-resolved transport physics to reactor-scale oxygen consumption and exposure heterogeneity without resolving individual cells in the reactor.

1 Goal, Scope and Scientific Advance

The central question is:

How does aggregate topology control oxygen uptake, oxygen-response dynamics and the fraction of oxygen-limited cells, and how can these effects be represented economically in reactor-scale models?

Current coarse models usually treat a biological aggregate as an equivalent homogeneous sphere. Its uptake is then calculated from size, bulk oxygen concentration and an empirical mass-transfer or effectiveness correlation. This can miss differences between aggregates that have the same size and porosity but different cell connectivity, shielding, surface accessibility and internal transport pathways.

The project will deliver three morphology-aware closures:

$$\eta_A = F_\eta(D_A, \lambda_A, \text{Pe}_A, \text{Da}_A), \quad (1)$$

$$\tau_A = F_\tau(D_A, \lambda_A, \text{Pe}_A, \text{Da}_A), \quad (2)$$

$$f_{\text{poor},A} = F_f(D_A, \lambda_A, \text{Pe}_A, \text{Da}_A, \chi_A), \quad (3)$$

where D_A is aggregate size, λ_A is a reduced morphology descriptor or class, Pe_A and Da_A describe transport and reaction, and χ_A is the reduced internal oxygen state. The effectiveness η_A determines equilibrium uptake, τ_A determines the memory of the internal oxygen state, and $f_{\text{poor},A}$ quantifies cellular exposure heterogeneity.

The resolved simulations concern isolated aggregates under controlled local conditions. They do not directly predict the spatial distribution of oxygen-limited cells in a reactor. Their role is to supply local constitutive relations. At reactor scale, the CFD solution provides the oxygen concentration and hydrodynamic conditions at the position of every population class or parcel. In this document, C_{bulk} denotes this *local unresolved carrier-phase concentration seen by the aggregate*; it is not the spatially averaged oxygen concentration of the entire reactor. The PBM or Euler–Lagrange model provides the local aggregate population. Evaluating the closures for all local classes or parcels then yields the reactor-wide distribution of uptake and oxygen-limited cell fractions.

2 State of the Art and Identified Gap

2.1 Continuum diffusion–reaction models

Established biofilm and granule models describe dissolved substrates through continuum diffusion–reaction balances coupled to microbial kinetics. The classical multispecies biofilm framework of Wanner and Gujer accounts for substrate gradients, external mass-transfer resistance and biomass development [1]. For suspended granules or pellets, the aggregate is commonly idealised as a homogeneous sphere with radial symmetry, an effective diffusivity and Monod-type reaction kinetics. The resulting one-dimensional problem predicts oxygen penetration depth, active-layer thickness or an effectiveness factor as functions of aggregate radius, effective diffusivity and kinetic parameters [2, 3].

These models are computationally efficient and remain the appropriate baseline. Diffusion is often the dominant internal transport mechanism, and a recent experimental and modelling study found that heterogeneous diffusivity had only a minor effect on the total flux for the investigated aerobic granules [4]. Consequently, the proposed project does not presuppose that a graph model is always necessary. It tests when the homogeneous-sphere approximation is sufficient and when it fails.

2.2 Evidence that structure can affect oxygen transport

Bioaggregates are not necessarily homogeneous continua. Microelectrode and imaging studies have shown dense cell clusters, voids, channels and spatially heterogeneous activity. De Beer et al. demonstrated that three-dimensional biofilm structure altered oxygen gradients and effective exchange area [5]. Transient measurements in aerobic granules further showed heterogeneous cell and extracellular-polymer distributions and structure-dependent oxygen response [6]. Reconstructed-geometry CFD studies subsequently connected granule structure with local hydrodynamics, penetration and reaction, obtaining information that is lost when the granule is treated as homogeneous [7]. A percolation model has likewise indicated that connected internal pores can enhance intragranular transport [8].

At the other end of the modelling spectrum, individual- and particle-based biofilm models resolve cells or biomass particles and couple them to diffusion–reaction fields [9, 10]. Such models can represent structural heterogeneity, but they are too expensive to evaluate for the complete aggregate population of a reactor-scale CFD calculation.

2.3 Data-driven structure–property reduction

Machine learning has been used to infer effective flow and transport properties of porous structures from geometrical descriptors or resolved images. For example, CFD-generated data have been mapped to permeability and filtration properties using fully connected and convolutional neural networks [11]. Graph networks are particularly suitable when the system is naturally represented by interacting particles or connected pores, because message passing preserves neighbourhood and connectivity information [13].

The unresolved gap is therefore not the calculation of oxygen diffusion in an idealised sphere, nor the cell-resolved simulation of one structure. It is the derivation of a *transferable and inexpensive closure* that retains only the topological information demonstrably needed for reactor-scale prediction. The novelty of the proposed project lies in combining particle-resolved flow–oxygen simulations, graph-based structural discovery and explicit distillation into CFD–PBM-compatible closures.

3 Hypotheses and Work-Package Logic

The scientific hypotheses are aligned directly with the work packages:

H1: Steady structure–transport relationship.

Aggregates with equal size and porosity can have different equilibrium uptake and poorly supplied cell fractions because topology changes shielding, accessibility and transport-path connectivity. WP2 tests H1.

H2: Morphology-dependent oxygen memory.

The response to changing external oxygen and flow conditions is not always instantaneous; its characteristic time depends on morphology and the competition among convection, diffusion and consumption. WP3 tests H2.

H3: Reducible topological information.

The structural information needed for H1 and H2 can be compressed into a small number of interpretable descriptors or morphology classes and used in inexpensive coarse-scale closures. WP4 tests H3.

WP1 is the enabling package that establishes the validated numerical framework and morphology library. Figure 1 shows the project logic.

4 Unified Physical and Numerical Framework

To avoid an unnecessarily large parameter space, the baseline system is defined once in Table 1. Approximately $N_A = 200$ spherical primary particles provide a useful starting point, but preliminary scaling and grid-cost tests will determine whether a range such as $N_A \in \{50, 200, 500\}$ is needed. The key design requirement is not a particular particle number; it is a morphology library containing structures with similar D_A and porosity but different topology.

4.1 Role of adhesion and distinction from the breakup project

Adhesion is necessary to create a physically meaningful contact network, but it is not a closure output in this project. The reference simulations use fixed structures so that the effect of topology

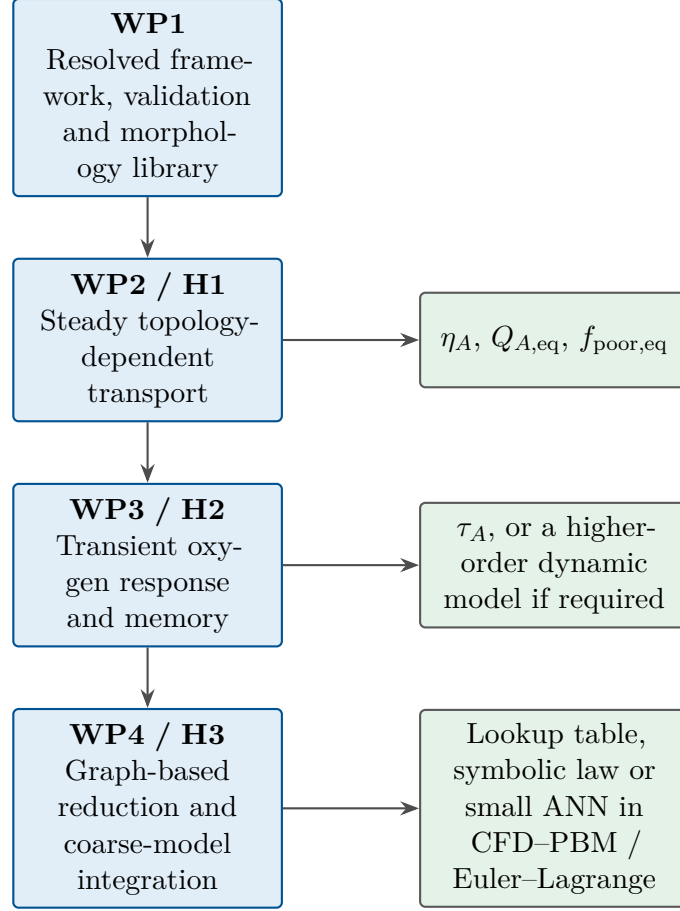


Figure 1: Work-package flow and direct correspondence between hypotheses and deliverables. WP2 and WP3 generate the physical closures; WP4 reduces and implements them.

Table 1: Compact definition of the baseline simulation framework.

Element	Baseline and controlled variations
Primary cells	Spherical, impermeable and approximately neutrally buoyant; no-slip momentum boundary and reactive oxygen boundary. Moderate size polydispersity can be added after the monodisperse reference case.
Aggregate state	Equivalent size D_A , particle number N_A , porosity, anisotropy, coordination/connectivity and surface accessibility. These quantities are stored as conventional descriptors and as a particle graph.
Mechanics	JKR or van der Waals adhesion is used to generate and stabilise plausible structures. The contact network is fixed during the transport simulations; the full aggregate may translate and rotate. Restructured or fragmented snapshots may be imported as prescribed morphologies, but their mechanics is not predicted here.
Flow	Single sphere and regular clusters for validation; uniform flow and simple shear for controlled analysis; homogeneous isotropic turbulence (HIT) for the main database; selected reactor-derived histories for transfer testing.
Transport and reaction	Conservative advection–diffusion with molecular diffusivity D_{O_2} . First-order surface uptake is the reference model; Monod uptake is introduced after transport validation.
Control variables	Pe_A , Da_A , D_A/η_K , aggregate orientation and selected turbulence parameters. A designed parameter study replaces a full Cartesian product.

on oxygen transport is identifiable. This choice separates the project from the running DFG project on fluid-induced breakup, which predicts breakup onset, mechanism and daughter sizes.

Restructuring and fragmentation need not be ignored. They can enter through a *one-way interface*: representative pre- and post-restructuring or fragment geometries are treated as members of the morphology library, and the new project quantifies their transport consequences. Thus the proposal can show that mechanics affects oxygen supply without duplicating the development of a breakup model. A later coarse reactor model may obtain morphological transitions from experiments, a PBM kernel or the existing breakup project.

4.2 Resolved flow and oxygen equations

The liquid is described by the incompressible Navier–Stokes equations. An immersed-boundary or fictitious-domain formulation resolves the no-slip surfaces of all primary cells. Dissolved oxygen satisfies

$$\frac{\partial C}{\partial t} + \mathbf{u} \cdot \nabla C = D_{\text{O}_2} \nabla^2 C + S_C. \quad (4)$$

At the surface S_i of cell i ,

$$-D_{\text{O}_2} \nabla C \cdot \mathbf{n} = j_i, \quad j_i = k_s C_{s,i}, \quad (5)$$

for the first-order baseline. A subsequent biological case uses

$$j_i = j_{\text{max}} \frac{C_{s,i}}{K_{\text{O}_2} + C_{s,i}}. \quad (6)$$

The surface-averaged concentration and uptake of each cell are

$$C_i(t) = \frac{1}{A_i} \int_{S_i} C \, dA, \quad Q_i(t) = \int_{S_i} j_i \, dA, \quad Q_A(t) = \sum_{i=1}^{N_A} Q_i(t). \quad (7)$$

These cell-level labels make it possible to compute total uptake and the internal distribution of oxygen exposure from the same simulation.

4.3 Dimensionless design space

The Péclet number measures convection relative to molecular diffusion:

$$\text{Pe}_A = \frac{U_A D_A}{D_{\text{O}_2}}, \quad \text{Pe}_\gamma = \frac{\dot{\gamma} D_A^2}{D_{\text{O}_2}} \quad \text{for shear-dominated conditions.} \quad (8)$$

It determines the external boundary layer and the extent to which flow can refresh internal pore spaces. The Damköhler number measures consumption relative to diffusive supply. For first-order uptake,

$$\text{Da}_A = \frac{k_s A_{\text{tot}}}{D_{\text{O}_2} D_A}. \quad (9)$$

Large Da_A favours strong internal depletion; large Pe_A can improve external and pore-scale supply, depending on topology. Expressing the database in terms of Pe_A and Da_A reduces dependence on one specific reactor or organism and makes the resulting closures transferable.

4.4 Validation and turbulence simulations

Validation proceeds from diffusion/reaction and forced convection around one sphere to regular clusters and then to uniform flow and simple shear. Conservation of total reactive flux, grid convergence, symmetry and available Sh limits are verified before HIT simulations.

The main database contains one structurally fixed aggregate per particle-resolved HIT simulation. A scalar-forcing term maintains a prescribed domain-mean bulk concentration despite oxygen consumption. Statistically stationary cases determine equilibrium quantities. Step, pulse and periodic changes of bulk oxygen, supplemented by a small number of reactor-derived histories, determine dynamic response. Different simulations represent the range of local conditions encountered along reactor trajectories; the reactor-scale population model later combines these local responses.

5 Graph Representation and Offline Learning

For aggregate r , the graph $\mathcal{G}_r = (\mathcal{V}_r, \mathcal{E}_r)$ contains one node per primary cell. Compact static node features describe radius, relative position, coordination, local void fraction, distance from the aggregate surface and accessible surface area. Two edge definitions are compared: physical contacts and spatial neighbours within a cutoff. Edge features describe relative position, gap/contact state and a local shielding or pore-throat measure. Global operating conditions such as Pe_A , Da_A and C_{bulk} are supplied separately.

The graph describes morphology; the resolved CFD fields generate targets. Dynamic variables such as $C_i(t)$ and $Q_i(t)$ are therefore primarily labels for learning and physical analysis, not indispensable inputs to the static graph encoder. This avoids hiding the answer inside the graph features.

5.1 GNN implementation in PyTorch Geometric

The GNN will be implemented in Python using *PyTorch Geometric* (PyG), a PyTorch-based framework for learning on graphs, point clouds and other irregular data structures [12]. Each aggregate is stored as one graph object with

$$\mathcal{G}_A = (\mathbf{X}_A, \mathbf{E}_A, \mathbf{A}_A, \mathbf{u}_A, \mathbf{y}_A), \quad (10)$$

where $\mathbf{X}_A$ contains particle/node features, $\mathbf{E}_A$ identifies the contact or neighbourhood edges, $\mathbf{A}_A$ contains edge features, $\mathbf{u}_A$ contains the separately supplied aggregate and operating variables, and $\mathbf{y}_A = [\eta_A, \tau_A, f_{\text{poor},A}]$ contains the resolved targets. Graphs with different particle numbers can be combined in mini-batches without padding them to one fixed size.

The initial architecture comprises three or four message-passing layers with edge features, followed by permutation-invariant global pooling. The pooled vector is the aggregate representation $\boldsymbol{\lambda}_A$. It is concatenated with D_A , N_A , Pe_A , Da_A and the local oxygen condition and passed to a small graph-level regressor. Edge-aware layers such as GINE are the primary option because relative distance, contact state and shielding are expected to matter; simpler graph-convolution and attention models provide baselines.

This computational strategy has precedent in related physical systems. Sanchez-Gonzalez et al. represented particles as nodes and local interactions as edges to learn the dynamics of fluids, rigid solids and deformable materials [13]. Graph representations have also been used to infer transport properties of pore networks from high-fidelity flow simulations [14]. These studies support graph-based learning for particle and transport systems, while the proposed morphology-aware oxygen closures for bioaggregates constitute the application-specific development.

5.2 Why use a GNN if the final closure is simple?

The GNN is an *offline discovery and teacher model*. It answers whether topology contains predictive information beyond hand-selected descriptors and which parts of the contact/neighbourhood

network matter. For each resolved aggregate,

$$\mathbf{z}_r = \mathcal{E}_\theta(\mathcal{G}_r), \quad \hat{\mathbf{y}}_r = \mathcal{H}_\theta(\mathbf{z}_r, D_{A,r}, \text{Pe}_{A,r}, \text{Da}_{A,r}), \quad (11)$$

where $\mathbf{y}$ contains η_A , τ_A and $f_{\text{poor},A}$, or the relevant subset for a given experiment. Attention/sensitivity analyses and comparison with conventional descriptors are used to interpret $\mathbf{z}_r$.

5.3 Separation of morphology and operating conditions

The graph encoder must describe the aggregate structure only. Therefore, for a fixed aggregate i ,

$$\boldsymbol{\lambda}_{A,i} = \mathcal{E}_\theta(\mathcal{G}_{A,i}) \quad (12)$$

is independent of the flow condition. The local operating conditions are introduced only in the subsequent prediction network:

$$\hat{\mathbf{y}}_{ij} = \mathcal{H}_\theta(\boldsymbol{\lambda}_{A,i}, D_{A,i}, N_{A,i}, \text{Pe}_{ij}, \text{Da}_{ij}, C_{\infty,ij}), \quad (13)$$

where j identifies the operating condition and

$$\mathbf{y}_{ij} = [\eta_{A,ij}, \tau_{A,ij}, f_{\text{poor},A,ij}]. \quad (14)$$

This architecture prevents flow information from being mixed into the morphology descriptor.

Consider ten fixed aggregate structures, each simulated under six different flow conditions. The resolved database then contains

$$10 \text{ structures} \times 6 \text{ conditions} = 60 \text{ labelled simulation cases.} \quad (15)$$

However, the graph encoder produces only

$$\boxed{10 \text{ unique morphology vectors}} \quad (16)$$

because each fixed graph is encoded once. The same vector $\boldsymbol{\lambda}_{A,i}$ is combined with six different sets of operating inputs:

$$\boldsymbol{\lambda}_{A,i}^{(1)} = \boldsymbol{\lambda}_{A,i}^{(2)} = \dots = \boldsymbol{\lambda}_{A,i}^{(6)}. \quad (17)$$

The six simulations teach the prediction network how one morphology responds to different environments; they do not define six different morphologies.

More generally, for N_G fixed graphs and N_C operating conditions, the database contains $N_G N_C$ labelled cases but only N_G unique morphology vectors. A separate morphology vector for each condition is appropriate only if flow causes deformation, restructuring or breakup, so that the underlying graph itself changes.

The complete graph and GNN are unavailable for every PBM class or parcel in a reactor. WP4 therefore compresses the teacher into a deployable morphology coordinate. This reduction is not an arbitrary assignment; it is constructed and validated in four explicit steps.

5.4 Derivation of an interpretable morphology coordinate

Step 1: construct candidate physical descriptors. For every aggregate in the resolved library, conventional and graph-derived descriptors are calculated before training. Candidate quantities include porosity ϕ_A , mean coordination number $\bar{Z}_A$, anisotropy a_A , accessible surface fraction $s_{\text{acc},A}$, mean shortest path from an internal cell to an accessible surface $\ell_{\text{surf},A}$, and

transport-network measures such as the size of the connected pore or neighbourhood component. They form

$$\mathbf{d}_A = [\phi_A, \bar{Z}_A, a_A, s_{\text{acc},A}, \ell_{\text{surf},A}, \dots]. \quad (18)$$

The set is intentionally compact and contains quantities that can ultimately be extracted from imaging, aggregate-generation simulations or empirical morphology distributions.

Step 2: discover missing structural information with the GNN. The GNN embedding $\mathbf{z}_A$ is compared with $\mathbf{d}_A$. Ablation tests answer whether size and porosity alone, then the full conventional descriptor vector, reproduce the resolved closure targets. The GNN is retained only if it reduces the prediction error on aggregates that were not used for training. Sensitivity analysis identifies which nodes, edges and descriptors drive this improvement.

Step 3: define the reduced coordinate or classes. Two alternatives are evaluated:

$$\lambda_A = g(\mathbf{d}_A), \quad \text{continuous descriptor}, \quad (19)$$

$$m_A = \mathcal{C}(\mathbf{z}_A) \in \{\text{M1, M2, M3}\}, \quad \text{discrete classes}. \quad (20)$$

The mapping g can be obtained by sparse regression or symbolic regression against the relevant GNN embedding coordinate and, more importantly, against the resolved transport targets. A purely illustrative result could be

$$\lambda_A \propto \frac{s_{\text{acc},A}}{(1 + \bar{Z}_A) \ell_{\text{surf},A}}, \quad (21)$$

where a high value represents an open, accessible structure and a low value a shielded structure with long internal paths. Equation (21) is not assumed beforehand; its functional form and necessity are determined from the data.

If discrete classes are more robust, clustering is performed in the validated embedding space and the clusters are translated into physical descriptions. An anticipated, but not prescribed, interpretation is given in Table 2.

Table 2: Illustrative interpretation of reduced morphology classes. The final thresholds and number of classes are determined from the resolved database.

Class	Structural interpretation	Expected transport tendency
M1: open	High accessible area, low shielding and short connected paths to the exterior	High effectiveness and small poorly supplied fraction
M2: intermediate	Moderate connectivity, accessibility and path length	Intermediate transport limitation
M3: shielded	Low accessible area, high coordination or long/tortuous internal paths	Low effectiveness and larger poorly supplied fraction

Step 4: distil and validate the deployable closure. The validated GNN teacher and the resolved cases are sampled over the required $(D_A, \lambda_A, \text{Pe}_A, \text{Da}_A)$ domain. A low-cost student is then fitted as either (i) a lookup table with interpolation, (ii) a symbolic correlation, or (iii) a small fully connected neural network if the dependence is too nonlinear for the first two forms. The student is accepted only if it remains accurate for withheld aggregate geometries and operating conditions.

The three deployment forms are alternatives, not three sequential ML models. The preferred order is symbolic correlation, lookup table and then small ANN, based on the balance of interpretability,

interpolation accuracy and numerical cost. A direct small ANN based only on prescribed bulk descriptors is retained as a baseline; the GNN is justified only if it adds demonstrable predictive information.

Symbolic regression is therefore a genuine alternative to the final small ANN, rather than an additional layer placed after it. Using, for example, PySR, candidate expressions are fitted directly to the resolved/GNN-teacher data,

$$\hat{y}_k = F_k(D_A, N_A, \lambda_A, \text{Pe}_A, \text{Da}_A, C_\infty), \quad y_k \in \{\eta_A, \tau_A, f_{\text{poor},A}\}, \quad (22)$$

while jointly considering prediction error and expression complexity. Physical constraints can restrict the admissible operators and enforce bounds such as $0 \leq \eta_A \leq 1$ and $0 \leq f_{\text{poor},A} \leq 1$. This route is especially well motivated here because Cranmer et al. demonstrated that symbolic regression can distil trained graph networks into explicit physical relations [15]; PySR provides a practical open-source implementation for scientific equation discovery [16]. If no compact expression reaches the prescribed validation accuracy, the small ANN is retained as the deployable closure.

5.5 Dynamic model and conditional role of Neural ODEs

The first dynamic model is deliberately simple. The aggregate carries a reduced oxygen or uptake state χ_A :

$$\frac{d\chi_A}{dt} = \frac{\chi_{A,\text{eq}}(C_{\text{bulk}}, \text{Pe}_A, \text{Da}_A, \lambda_A) - \chi_A}{\tau_A(D_A, \lambda_A, \text{Pe}_A, \text{Da}_A)}. \quad (23)$$

The response time is obtained from resolved step responses, for example using the integral of the normalised relaxation curve. This one-state equation is the primary deliverable because it is interpretable and inexpensive.

Only if step, pulse and periodic tests show systematic multi-time-scale or hysteretic behaviour will a graph-conditioned Neural ODE be investigated:

$$\frac{d\chi_A}{dt} = \mathbf{F}_\theta(\chi_A, C_{\text{bulk}}, \text{Pe}_A, \text{Da}_A, \lambda_A). \quad (24)$$

The Neural ODE does not automatically become a lookup table. It is either retained as a small dynamic operator or distilled into the lowest-order multi-state ODE that reproduces the resolved histories. This decision is made from validation evidence, not fixed in advance.

6 Closure Definitions and Use in the Coarse Model

6.1 Equilibrium uptake

For a known biological kinetic law evaluated at local bulk oxygen,

$$Q_{\text{kin}} = N_A q(C_{\text{bulk}}), \quad Q_{A,\text{eq}} = \eta_A Q_{\text{kin}}. \quad (25)$$

For example,

$$q(C_{\text{bulk}}) = q_{\text{max}} \frac{C_{\text{bulk}}}{K_{\text{O}_2} + C_{\text{bulk}}}. \quad (26)$$

The closure in Eq. (1) corrects the hypothetical fully supplied rate for transport limitations caused by boundary layers, shielding and internal depletion.

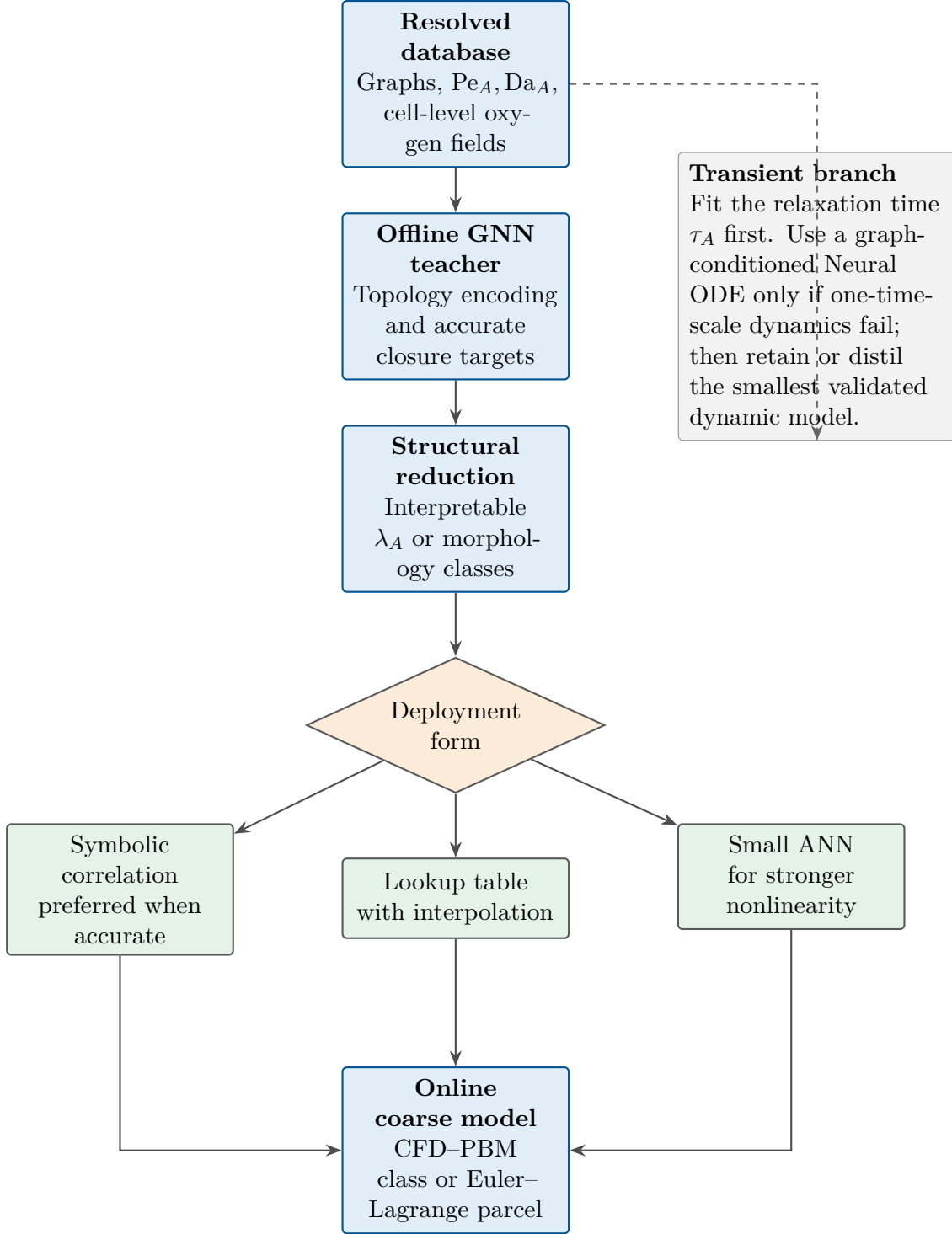


Figure 2: Machine-learning and model-reduction procedure. The GNN is an offline teacher; only one selected low-cost student closure is evaluated online. Neural ODE modelling is a conditional transient branch and is not required for the equilibrium closure.

6.2 Poorly supplied cell fraction

For a defined biological threshold C_{crit} , the resolved target is

$$f_{\text{poor},A}(t) = \frac{1}{N_A} \sum_{i=1}^{N_A} \mathbb{I}[C_i(t) < C_{\text{crit}}]. \quad (27)$$

At equilibrium this is a direct closure of morphology and local conditions. Under transients, it additionally depends on the carried internal state χ_A . It is not an oxygen sink by itself; it is an exposure indicator available to models of growth, productivity, viability or metabolic state.

6.3 Local quantities seen by a PBM class or parcel

For an Euler–Lagrange parcel p , the oxygen concentration supplied to the closure is interpolated from the carrier-phase field:

$$C_{\infty,p}(t) = C(\mathbf{x}_p(t), t). \quad (28)$$

This is the coarse-grid concentration surrounding the unresolved aggregate. It plays the role of C_{bulk} in the particle-resolved training case, but it is neither a reactor average nor the unresolved concentration inside the aggregate. The local relative velocity and shear magnitude are, for example,

$$U_{\text{rel},p} = |\mathbf{u}(\mathbf{x}_p, t) - \mathbf{v}_p|, \quad (29)$$

$$\dot{\gamma}_p = \sqrt{2\mathbf{S} : \mathbf{S}}|_{\mathbf{x}_p}, \quad \mathbf{S} = \frac{1}{2} (\nabla \mathbf{u} + \nabla \mathbf{u}^T). \quad (30)$$

Accordingly, the relevant Péclet number can be based on slip or shear:

$$\text{Pe}_{\text{slip},p} = \frac{U_{\text{rel},p} D_{A,p}}{D_{\text{O}_2}}, \quad \text{Pe}_{\text{shear},p} = \frac{\dot{\gamma}_p D_{A,p}^2}{D_{\text{O}_2}}. \quad (31)$$

The closure development determines whether one of these definitions or both are required. An Eulerian CFD–PBM uses the corresponding local CFD values for each size–morphology class in a control volume.

The final online closure therefore combines a morphology property carried by the parcel with aggregate properties and locally evaluated reactor conditions:

$$\boxed{\begin{bmatrix} \eta_{A,p} \\ \tau_{A,p} \\ f_{\text{poor},A,p} \end{bmatrix}} = \mathbf{F}(D_{A,p}, N_{A,p}, \boldsymbol{\lambda}_{A,p}, \text{Pe}_p, \text{Da}_p, \chi_p). \quad (32)$$

Here $\boldsymbol{\lambda}_{A,p}$ originates from the offline graph analysis, whereas Pe_p , Da_p and the surrounding oxygen concentration are calculated from the local reactor state. In an Euler–Lagrange implementation, the fluid quantities are interpolated at the parcel position $\mathbf{x}_p$, representing the centre of the unresolved aggregate. They are not values inside a particle-resolved sphere. For aggregates that are large relative to the CFD grid, a kernel-weighted average over neighbouring cells may be used instead of single-point interpolation.

The online reactor model does not evaluate the complete GNN. It stores only $\boldsymbol{\lambda}_{A,p}$, assigned at parcel creation from measured descriptors, an aggregate-generation model or a prescribed conditional distribution. The complete separation is

$$\underbrace{\mathcal{G}_{A,i} \longrightarrow \boldsymbol{\lambda}_{A,i}}_{\text{offline graph encoder}} \quad \underbrace{(D_A, N_A, \boldsymbol{\lambda}_A, \text{Pe}_A, \text{Da}_A, \chi_A) \longrightarrow (\eta_A, \tau_A, f_{\text{poor},A})}_{\text{online low-cost closure}}. \quad (33)$$

6.4 Online calculation for one PBM class or parcel

The final model performs the following inexpensive calculation at each reactor location and time:

1. Interpolate the local carrier-phase oxygen concentration $C_{\infty,p}$, velocity gradients, turbulence quantities and fluid properties from CFD.
2. Read class/parcel properties D_A , N_A and λ_A .
3. Calculate Pe_A and Da_A .
4. Query the selected lookup table, symbolic correlation or small ANN for η_A , τ_A and $f_{\text{poor},A}$.
5. Calculate $Q_{A,\text{eq}} = \eta_A N_A q(C_{\infty,p})$, update the dynamic state with Eq. (23), and obtain current Q_A and $f_{\text{poor},A}$.

For a morphology-resolved PBM,

$$n = n(D_A, \lambda_A, \mathbf{x}, t), \quad S_{\text{O}_2}(\mathbf{x}, t) = - \sum_{k,m} n_{k,m} Q_{A,k,m}. \quad (34)$$

The same logic applies to an Euler–Lagrange parcel with statistical weight w_p , whose sink contribution is $-w_p Q_{A,p}$. The local number or fraction of poorly supplied cells is reconstructed by weighting $f_{\text{poor},k,m}$ with $N_{A,k} n_{k,m}$, or the corresponding parcel weights. The reactor distribution therefore emerges from many local evaluations of single-aggregate closures, not from placing individual cells in the reactor mesh.

If morphology is resolved only by discrete classes, the PBM carries a small number of size–morphology populations. If only a measured conditional morphology distribution $P(\lambda_A \mid D_A)$ is available, the closures can be averaged over that distribution, with a documented loss of morphology-specific information.

6.5 How morphology is assigned in the reactor model

The transport closure cannot infer the morphology of an unresolved reactor parcel from local oxygen and shear alone. A morphology coordinate must therefore enter as a parcel property or PBM internal coordinate. Three levels of implementation are distinguished.

Level 1: prescribed morphology distribution. For the first reactor demonstrator, a parcel receives its morphology at injection or creation from a measured or simulated conditional distribution,

$$\lambda_p \sim P(\lambda_A \mid D_{A,p}), \quad \text{or} \quad m_p \sim P(m_A \mid D_{A,p}). \quad (35)$$

It then carries

$$\{D_{A,p}, N_{A,p}, \lambda_p, \chi_p, w_p\}, \quad (36)$$

and λ_p remains fixed along the trajectory. This is the reference assumption for demonstrating the oxygen closures without introducing an unvalidated morphology-evolution model.

Level 2: descriptor-based assignment. If imaging or an aggregate-generation model supplies porosity, coordination, anisotropy and accessibility, Eq. (19) is evaluated directly when the parcel is created. The coarse simulation stores only λ_p , not the complete graph.

Level 3: morphology-resolved population dynamics. An extended PBM uses $n(D_A, \lambda_A, \mathbf{x}, t)$ and includes transition kernels for aggregation, growth, restructuring or breakup. These kernels determine how a new or transformed aggregate enters another morphology class. Their derivation is not part of the core oxygen-transport project. They may be supplied by experiments, aggregate-generation simulations or the existing breakup project.

6.6 Explicit assumptions of the first coarse-scale demonstrator

The proposed first implementation therefore assumes that:

1. every parcel represents aggregates with one size and one reduced morphology value or class;
2. the morphology distribution at parcel creation is known or prescribed;
3. morphology remains fixed during a parcel trajectory unless an external transition model is supplied;
4. the local CFD concentration $C_{\infty,p}$ is resolved on the reactor mesh, whereas internal aggregate oxygen is represented only by χ_p ;
5. the oxygen closure predicts uptake and exposure, not aggregate growth or morphology evolution;
6. a separate biological law is required to convert oxygen exposure into a PBM growth rate.

These assumptions make the final use of the model unambiguous. The core deliverable is a morphology-conditioned oxygen model. A morphology-evolution model and an oxygen-dependent growth kernel are downstream couplings, not quantities silently inferred by the GNN.

7 Work Packages, Milestones and Outputs

WP1: Resolved framework, validation and morphology library

WP1 defines a reference biological system, implements conservative reactive transport in the particle-resolved solver and validates single particles, regular clusters and canonical flows. It constructs the controlled morphology library and establishes the feasible ranges of N_A , Pe_A and Da_A . **Milestone M1:** verified mass conservation, grid convergence and a database design that exhibits measurable transport limitation.

WP2: Steady topology-dependent transport (H1)

WP2 performs steady uniform-flow, shear and HIT simulations across the designed structure–transport space. Paired structures with equal size and similar porosity isolate the effect of topology. The outputs are η_A , $Q_{A,eq}$, equilibrium $f_{poor,A}$, and a mechanistic analysis of shielding, accessibility and flow penetration. **Milestone M2:** quantitative acceptance or rejection of H1 and validated steady targets for model reduction.

WP3: Transient oxygen response (H2)

WP3 applies step, pulse, periodic and selected reactor-derived oxygen histories to representative morphologies. It determines τ_A , tests the validity of the one-state relaxation model and introduces a higher-order ODE or Neural ODE only when justified. **Milestone M3:** a validated dynamic model over the relevant ratio of aggregate and reactor fluctuation time scales.

WP4: Graph reduction and reactor-scale integration (H3)

WP4 trains and interprets the GNN teacher, constructs λ_A or discrete morphology classes, and distills the teacher into the simplest accurate online closure. The three closures are implemented first in an Euler–Lagrange demonstrator and, subject to feasibility, in a morphology-resolved CFD–PBM. **Milestone M4:** reactor-scale predictions of oxygen sink and oxygen-limited cell fractions with quantified closure uncertainty.

The main outputs are therefore:

1. a validated particle-resolved solver and database for reactive transport in bioaggregates;
2. physical evidence for when topology matters beyond size and porosity;
3. closures for equilibrium effectiveness, oxygen-response time and poorly supplied cell fraction;
4. an interpretable morphology representation discovered with graph-based learning;
5. a verified low-cost closure implementation for CFD–PBM or Euler–Lagrange reactor simulation.

8 Feasibility, Risk Control and Non-Overlap

The decisive preliminary test is whether realistic size and uptake ranges produce effects large enough to matter:

$$\eta_A < 1, \quad f_{\text{poor},A} > 0, \quad \tau_A/\tau_{\text{reactor}} \not\ll 1, \quad (37)$$

and whether equal-size, equal-porosity structures differ beyond numerical and sampling uncertainty. A diffusion estimate $\tau_D \sim R_A^2/D_{\text{eff}}$ and a small resolved pilot matrix will determine the aggregate-size and kinetic ranges before the full database is generated.

The project is modular if individual assumptions fail. If topology adds no information beyond conventional descriptors, that negative result supports a simpler closure without a morphology coordinate. If a single response time is sufficient, the optional Neural ODE is omitted. If morphology-resolved PBM transport is too expensive or lacks transition data, the closures are first demonstrated with Euler–Lagrange parcels or morphology-averaged PBM classes.

Non-overlap with the running DFG project is defined by the predicted quantities. The existing project studies mechanical breakup onset, mechanism, fragment motion and daughter-size distributions and derives a breakup closure. The proposed project predicts reactive mass transfer, oxygen memory and cell-level oxygen exposure for prescribed morphologies. Shared numerical infrastructure and selected aggregate geometries constitute enabling preliminary work; the new simulations, labels, hypotheses and coarse-scale closures concern oxygen transport rather than mechanical failure.

9 Concise Project Statement

The project will use particle-resolved DNS–DEM simulations to determine how bioaggregate topology controls equilibrium oxygen uptake, transient oxygen memory and the fraction of poorly supplied cells. A GNN will serve offline to discover and encode the necessary structural information. Its knowledge will then be distilled into one inexpensive deployable form—a symbolic correlation, lookup table or small neural network—which is evaluated for aggregate classes or parcels in a coarse CFD–PBM or Euler–Lagrange reactor model. The resulting framework will predict reactor-scale oxygen demand and the spatial distribution of oxygen-limited cellular material without resolving individual cells in the reactor.

10 Preliminary DEM and graph-based analysis

10.1 Agglomerate generation and numerical design

The preliminary data set consists of 135 monodisperse agglomerates generated with the local-filling-factor algorithm of Ringl and Urbassek. In this method, a new primary particle is attached preferentially to a particle with a low local filling factor. The detector radius is written as

$$r_\sigma = m_\sigma r_p, \quad (38)$$

where r_p is the primary-particle radius and m_σ is the detector-radius multiplier. The latter controls the generated morphology but is not itself the fractal dimension. Consequently, the requested values of D_f should be regarded as design labels until the resulting structures are independently evaluated, for example by box counting or a radius-of-gyration method.

The numerical design was

$$\begin{aligned} & 3 \text{ particle diameters} \times 3 \text{ particle counts} \\ & \times 3 \text{ requested fractal dimensions} \times 5 \text{ random realizations} = 135 \text{ cases.} \end{aligned} \quad (39)$$

Design variable	Values
Primary-particle diameter d_p	1.0, 1.5, 2.0 (simulation length units)
Number of particles N	50, 100, 200
Requested fractal dimension D_f	1.8, 2.2, 2.6
Random realizations	5 per parameter combination
Total number of cases	135

Each realization was assigned a unique random seed. The generated particle coordinates and radii were exported as CSV, LIGGGHTS data (.sys), and VTK files. The LIGGGHTS data contain one granular atom type, no bonds, zero initial translational and angular velocities, and an initial computational box from -100 to 100 in each coordinate direction.

10.2 DEM relaxation and material parameters

The generated structures were relaxed with LIGGGHTS using adhesive particle contacts. Particle and contact data were stored at multiple time steps up to step 100000. The particle output contains positions, radii, particle IDs and velocities, while the contact output contains the interacting particle IDs, gap, overlap, contact area, contact state, normal and tangential forces, total force and contact torque. These data were converted to VTK for inspection in ParaView and subsequent graph construction.

The parameters confirmed directly from the generated particle data are listed below. The remaining values must be copied from the final LIGGGHTS input before this text is used in a report or publication.

Property	Value used
Particle shape	Sphere
Particle diameter d_p	1.0, 1.5, 2.0 (simulation length units)
Particle density ρ_p	2.0 (in the selected LIGGGHTS unit system)
Initial translational/angular velocity	Zero
Bond types	None
Contact model	[insert model from LIGGGHTS input]
Young's modulus E	[insert verified value and unit]
Poisson ratio ν	[insert verified value]
Coefficient of restitution e	[insert verified value]
Sliding-friction coefficient μ	[insert verified value]
Rolling-friction coefficient μ_r	[insert verified value]
Hamaker constant A_H	[insert verified value in J]
Minimum separation $h_{\min}$	[insert verified value and unit]
Time step Δt	[insert verified value and unit]

For each case, particle and contact VTK files were matched by time-step number. A snapshot was classified as relaxed after three consecutive output intervals satisfied both

$$\frac{\Delta x_{\text{RMS}}}{d_p} \leq 10^{-4}, \quad 1 - \frac{|C_k \cap C_{k+1}|}{|C_k \cup C_{k+1}|} \leq 0.01, \quad (40)$$

where C_k denotes the set of particle contacts at time step k . If no snapshot fulfilled these criteria, the last common particle/contact time step was retained and marked as non-converged.

10.3 Graph construction and supervised targets

Each relaxed agglomerate was represented as a graph. Primary particles form the nodes and physical particle contacts form the edges. Every physical contact was stored in both directions for message passing. Thus, an agglomerate with N particles and N_c contacts has N nodes and $2N_c$ directed edges.

The preprocessing stored three node features, but the minimal GNN used only two structural node scalars:

$$\mathbf{x}_i = \left[\frac{|\mathbf{x}_i - \mathbf{x}_c|}{R_g}, z_i \right], \quad (41)$$

where d_{ref} is twice the median particle radius, $\mathbf{x}_c$ is the agglomerate center, R_g is the radius of gyration, and z_i is the coordination number. Normalized, centered particle coordinates were supplied separately to the encoder. The stored radius ratio was omitted because it is approximately constant for the present monodisperse particles.

The converted DEM data provide 22 available edge/contact features, including distance, overlap, gap, contact area, force and torque quantities and contact flags. For the minimal model, only the normalized centre distance d_{ij}/d_{ref} was selected as a stored edge scalar, because connectivity and interparticle geometry are the quantities directly relevant to the three purely structural targets considered here. Graph connectivity and the normalized relative-position vector were additionally supplied for message passing. Force, torque, overlap and contact-area fields were retained in the processed data for sensitivity studies, but were not used by the reported minimal model.

The convex hull of the particle centers was calculated with a small numerical perturbation (Qhull option QJ). Hull vertices were classified as surface particles. A breadth-first search through the contact graph then defined the graph depth h_i of every particle from this surface. Three graph-level

proxy targets were calculated:

$$\bar{h} = \frac{1}{N} \sum_{i=1}^N h_i, \quad (42)$$

$$f_{h \geq 2} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(h_i \geq 2), \quad (43)$$

$$S_{\text{access}} = \frac{1}{N} \sum_{i=1}^N \exp(-\beta h_i), \quad \beta = 0.5. \quad (44)$$

These quantities represent geometric shielding and accessibility proxies. They are not yet oxygen-transport or oxygen-uptake results.

10.4 GNN architecture and preliminary training

The model embeds the two selected node scalars together with normalized particle coordinates, and the selected edge-distance scalar together with relative-position vectors, into a hidden space of dimension 64. Six residual GINE message-passing layers with layer normalization and dropout are applied. Global mean, maximum and size-normalized additive pooling produce one aggregate-level representation, which is compressed to a latent vector

$$\lambda_A = [\lambda_1]. \quad (45)$$

Latent dimensions from one to four were tested. The selected dimension was the smallest candidate whose mean out-of-fold R^2 was within 0.02 of the best candidate. This rule selected one latent coordinate. A final two-layer prediction head maps λ_1 to $\bar{h}$, $f_{h \geq 2}$ and S_{access} ; no precomputed aggregate-level descriptor is supplied to the prediction head.

Training used Adam with a learning rate of 10^{-3} , weight decay of 10^{-5} , at most 300 epochs, batch size 16, dropout 0.1 and random seed 7. Validation-based early stopping and random rigid-rotation augmentation were used. Targets were standardized using only the corresponding training data. All 135 graphs were processed successfully. Five-fold cross-validation was grouped by parameter combination, so that all random realizations of one (d_p, N, D_f) combination remained in the same held-out fold. The selected one-dimensional model gave:

Fold	RMSE($\bar{h}$)	RMSE($f_{h \geq 2}$)	RMSE(S_{access})
1	0.315	0.051	0.044
2	0.345	0.053	0.036
3	0.452	0.054	0.041
4	0.448	0.047	0.031
5	0.393	0.040	0.033
Mean	0.390	0.049	0.037

The combined out-of-fold R^2 values were 0.869 for mean graph depth, 0.894 for shielded fraction and 0.913 for accessibility score, with corresponding MAE values of 0.293, 0.039 and 0.030. The mean R^2 across the three targets was 0.892. Latent dimensions two, three and four gave mean R^2 values of 0.887, 0.888 and 0.876, respectively; hence additional latent coordinates did not improve the result.

Particle-count-only baselines achieved mean out-of-fold R^2 values between 0.754 and 0.776, showing that the graph model captures structural variation beyond particle count alone. A stricter baseline used particle count, radius of gyration, contact count, contact density, mean coordination and maximum coordination. Linear regression with these conventional descriptors achieved a mean

R^2 of 0.891, while a random forest achieved 0.909, slightly exceeding the GNN by 0.016. Thus, the present proxy-target study demonstrates morphology information beyond particle count, but it does not yet establish that detailed graph encoding is necessary beyond well-chosen scalar descriptors. The more decisive test will use resolved oxygen-transport targets for aggregates with similar conventional descriptors but different internal topology.

The present results demonstrate the complete DEM-to-graph-to-GNN workflow, but remain preliminary. The cross-validation latent values originate from independently trained fold models and are therefore not interpreted as one common latent coordinate. For descriptive latent analysis, final models are trained on all graphs and their arbitrary sign and scale are aligned across random seeds. Such all-data latent values are used only for interpretation, not as leakage-free evidence of predictive performance.

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